In everyday life in the modern universe, each of the fundamental forces of nature has a distinct role. Gravity holds us to the ground, electricity keeps our lights on, magnetism holds our shopping list to the fridge, the weak nuclear force makes sure our backyard nuclear reactor keeps glowing a nice steady blue, and the strong nuclear force prevents our bodies' protons and neutrons from decomposing into their component parts. But the physical laws governing how these forces work, how they interact with each other, and even whether or not it's possible to tell them apart, depend on the conditions under which they're measured. Specifically, the ambient energy, or temperature. At high enough energies, the forces start to merge and combine, rearranging the structure of particle interactions and the laws of physics themselves.

It's been known for some time that, even under everyday circumstances, electricity and magnetism are aspects of the same phenomenon, which is why electromagnets are a thing and why dynamos can generate electricity. This kind of unification is like candy to physicists. Any time we can take two complex phenomena and say, "Actually, when you look at it *this* way, they're THE SAME THING," we basically explode with physics joy. In some ways, this is the ultimate goal of theoretical physics: to find a way to take all the complicated messy stuff we see around us and rearrange it into something pretty and compact and simple that just *looks* complicated because of our weird low-energy perspective.

Where the forces of particle physics are concerned, this quest is called Grand Unification. Based on theory and extrapolations from what we see in lab experiments, it's thought that at very high energies, electromagnetism, the weak force, and the strong force all come together to be something else entirely, such that there's no way to distinguish them—they're all part of the same kind of big particle-energy mix governed by a Grand Unified Theory. There have been a few GUTs developed and proposed, but the difficulty of accessing the energy scales where the unification occurs makes them hard to confirm or rule out, so we'll just call this "an area of active research" that would very much appreciate any more funding you'd like to throw at it.

You might notice that gravity is not invited to the GUT party. To bring gravity into the picture, we need something grander and more unifying than even a Grand Unified Theory—we need a Theory of Everything (aka TOE). There's a general belief among physicists that sometime around the Planck Time, gravity *was* somehow unified with the other forces (along with the